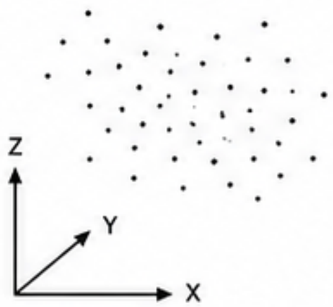
XMST with Percolation–Jenks Fracturing: Method Overview

From stellar catalogue to recovered stellar structures

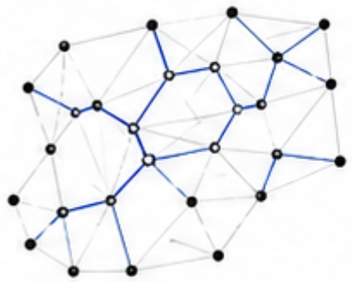
1 Input: Stellar Catalogue

Heliocentric Cartesian positions (X, Y, Z), derived from Galactic longitude, latitude and distance, from the Gaia DR3 OB-star catalogue.



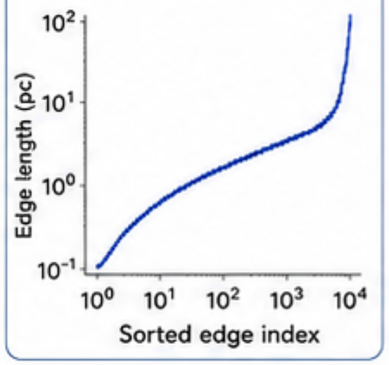
2 Coordinates, Delaunay Graph & Global MST

Use catalogue heliocentric Cartesian coordinates (Sun at origin). Construct a 3D Delaunay triangulation, then compute the global Minimum Spanning Tree (Kruskal).



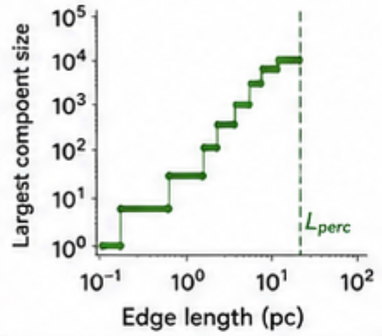
3 MST Edge-Length Distribution

Calculate and sort all global MST edge lengths.



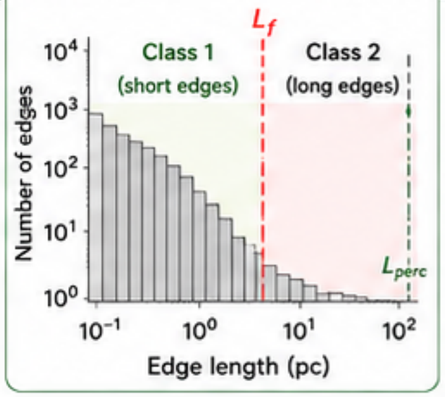
4 Percolation Analysis

Build the MST from shortest to longest edges. Record the size of the largest connected component. The largest jump defines the critical transition edge.



5 Jenks Natural Breaks

Apply two-class Jenks Natural Breaks to the MST edge-length distribution using only edges at or below the percolation limit.



6 Select Fracture Scale

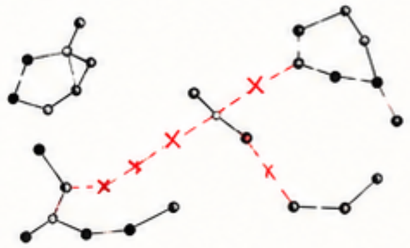
The Jenks class boundary is adopted as the fracture scale, L_f, which defines the XMST segmentation threshold.

All MST edges longer than L_f are removed.

Fracture scale L_f

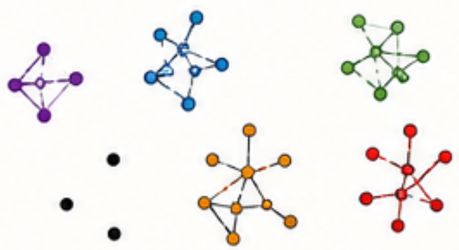
7 Fracture the MST

Remove all edges with length > L_f. Recovered structures remain as disconnected subtrees.



8 Identify Candidate Groups

Label each connected component with at least N_{min} members as a candidate structure. Discard smaller groups and isolated nodes.



9 Validate & Measure

For each recovered group, compare with injected structures using completeness, purity and recovery statistics. Assess membership stability under distance perturbations.

Completeness $= \frac{N_{correct}}{N_{injected}}$



Purity $= \frac{N_{correct}}{N_{recovered}}$



Recovery (Jaccard) $= \frac{|A \cap B|}{|A \cup B|}$



Stability across distance perturbations



10 Outputs

Catalogue of recovered structures with key properties, including membership lists, fracture scale, purity, completeness and stability metrics.

ID	N	Radius (pc)	L_f (pc)	Purity	Completeness	σ_d (pc)
1	23	2.46	17.02	0.77	0.88	32
2	15	3.89	17.03	0.68	0.75	28
3	34	5.62	17.01	0.83	0.81	35
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Key Features of XMST

- Objective fracture-scale selection
- Works on irregular stellar structures
- Stable under uncertainty propagation
- Produces a clean candidate-structure catalogue

Notation

MST	Minimum Spanning Tree
CDF	Cumulative Distribution Function *(comparison method only)*
L_f	Jenks fracture scale (critical edge length)
N_{min}	Minimum group size (candidate threshold)
pc	Parsecs